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PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

B. Lasers

time limit per test: 2 seconds
memory limit per test: 256 megabytes

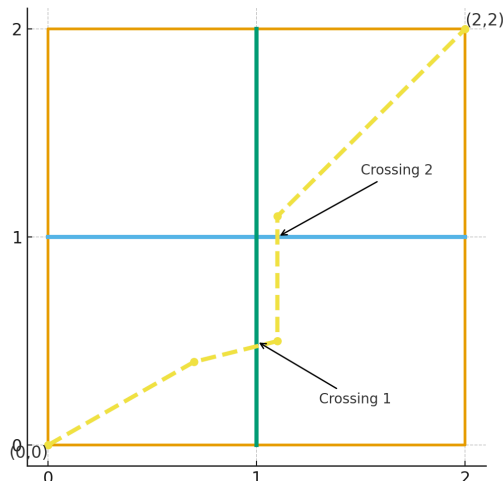
There is a 2D-coordinate plane that ranges from $(0, 0)$ to (x, y) . You are located at $(0, 0)$ and want to head to (x, y) .

However, there are n horizontal lasers, with the i -th laser continuously spanning $(0, a_i)$ to (x, a_i) . Additionally, there are also m vertical lasers, with the i -th laser continuously spanning $(b_i, 0)$ to (b_i, y) .

You may move in any direction to reach (x, y) , but your movement must be a continuous curve that lies inside the plane. Every time you cross a vertical or a horizontal laser, it counts as one crossing. Particularly, if you pass through an intersection point between two lasers, it counts as **two crossings**.

For example, if $x = y = 2$, $n = m = 1$, $a = [1]$, $b = [1]$, the movement can be as follows:

Minimum crossings = 2 (must cross $x=1$ and $y=1$ once each)



What is the minimum number of crossings necessary to reach (x, y) ?

Input

The first line contains t ($1 \leq t \leq 10^4$) — the number of test cases.

The first line of each test case contains four integers n , m , x , and y ($1 \leq n, m \leq 2 \cdot 10^5$, $2 \leq x, y \leq 10^9$).

The following line contains n integers $a_1, a_2, \dots, a_n$ ($0 < a_i < y$) — the y-coordinates of the horizontal lasers. It is guaranteed that $a_i > a_{i-1}$ for all $i > 1$.

The following line contains m integers $b_1, b_2, \dots, b_m$ ($0 < b_i < x$) — the x-coordinates of the vertical lasers. It is guaranteed that $b_i > b_{i-1}$ for all $i > 1$.

It is guaranteed that the sum of n and m over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output the minimum number of crossings necessary to reach (x, y) .

Example

input	Copy
2	
1 1 2 2	

Codeforces Round 1050 (Div. 4)

Contest is running

01:16:57

Contestant



→ Submit?

Language: Python 3.13.2
Almost always, if you send a solution on PyPy, it works much faster

Choose file: Choose File No file chosen

Submit

```
1
1
2 1 100000 100000
42 58
32
```

output

Copy

```
2
3
```

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